A Quantity Theory of Money Approach to Cryptocurrency Valuation: Store of Value, Medium of Exchange, and the Economic Moat Signal

Moazzam Khoja

University of Texas at San Antonio

moazzam.khoja@utsa.edu

June 2026

Preliminary. Comments welcome.

Generated with AI assistance under AFA 2027 Special Session guidelines.

Abstract

We apply the Quantity Theory of Money (QTM) to the cross-sectional valuation of cryptocurrencies and protocol tokens. Treating each digital asset as the currency of a self-contained economy, we decompose market capitalization into a Medium of Exchange (MoE) component, priced by the free-circulating velocity of supply, and a Store of Value (SoV) residual. The key innovation is λ—the fraction of total supply that is locked or staked—which partitions the circulating supply into economically active and inert components. A central result is that the SoV/MoE ratio simplifies to $\lambda/(1-\lambda)$, directly observable from on-chain data without velocity estimation. For native blockchain coins, high λ reflects a convenience yield and seigniorage signal grounded in monetary

search theory. For governance and utility tokens, high λ is an economic moat signal, capturing token holder fundamental conviction in platform growth. We show that $\lambda/(1-\lambda)$ alone positively predicts future cross-sectional returns, and that this predictability is substantially stronger when the asset is cheap relative to its fundamental economic activity—measured by our NVT/fundamental ratio—supporting the interpretation that NVT identifies when the conviction signal is not yet priced in. A long Stars/short Avoid portfolio generates positive factor-adjusted alpha after controlling for market, size, and momentum factors.

1 Introduction

What determines the value of a cryptocurrency? The dominant approach in the literature has been to adapt discounted cash flow (DCF) models, searching for a dividend-like flow—transaction fees, staking rewards, governance rights—that can be capitalized into a present value (Liu and Tsyvinski, 2021; Pagnotta and Buraschi, 2022; Biais, Bisière, Bouvard, and Casamatta, 2019). This approach, however, mischaracterizes the fundamental nature of digital assets. Cryptocurrencies and protocol tokens exhibit currency-like behavior: each coin or token is the monetary instrument of a self-contained digital economy, and its value is governed by the same forces that determine the value of any currency—the scale of economic activity it enables and the efficiency with which it circulates. The correct framework, we argue, is the Quantity Theory of Money (Fisher, 1911).

The Fisher equation, $MV = PQ$, relates the stock of money (M) and its velocity (V) to the nominal value of economic output (PQ) (Fisher, 1911). Applied to digital assets, a cryptocurrency's market capitalization proxies its money supply, and the economic output it supports—on-chain transaction volume for native coins, protocol throughput for tokens—is its PQ. This is not merely an analogy. Cong, Li, and Wang (2021) develop a formal dynamic model in which token prices aggregate heterogeneous users' transactional demand, with equilibrium prices governed by platform adoption—a formulation directly consistent with the QTM framework we develop here. Sockin and Xiong (2020) similarly model tokens as platform membership instruments, showing that the token price reflects the present value of the convenience yield accruing to participants: the non-pecuniary benefit of network membership, including access to platform services, network liquidity, and future transactional capability.

Before proceeding, we define the two components of value our framework decomposes. The *Medium of Exchange* (MoE) value of a digital asset is the value attributable to its current transactional function: the economic activity it enables today, priced by the velocity with which it circulates. The *Store of Value* (SoV) value is the premium agents are willing to pay today by forgoing current exchange utility—the transactions, goods, services, or alternative investments they could obtain by spending the asset now—in anticipation of future value appreciation. This appreciation is expected to arise from network growth, expanding user adoption, increasing utility, and innovation. An agent holding Ethereum rather than spending it is implicitly paying an intertemporal price: she sacrifices today's exchange utility for her conviction that the network's future worth exceeds today's use value. The SoV component is the aggregate market valuation of this forward-looking conviction.

The QTM framework requires a critical extension to operationalize this decomposition. Not all of a cryptocurrency's circulating supply is economically active. A substantial and growing fraction is locked in staking contracts, pledged as governance collateral, or held in long-term custody with near-zero velocity. These holdings are economically inert from a transactional standpoint: they contribute to the money supply but perform no current exchange function. Conflating active and inactive supply distorts velocity estimates and, consequently, misprices the economic content of market capitalization. We address this by introducing λ, defined as the proportion of the total supply that is locked or staked—including tokens committed to network validation, governance contracts, and other long-term holding functions. The formal derivations are provided in Appendix A. The key result is that the SoV/MoE ratio simplifies to $\lambda/(1-\lambda)$: a monotone, directly observable function of λ alone, requiring no velocity estimation.

Why does λ matter beyond its role in the decomposition? For native blockchain coins—Ethereum, Solana, and their peers—high λ is grounded in three complementary theoretical traditions. First, *monetary search theory* (Kiyotaki and Wright, 1993; Lagos and Wright, 2005) establishes that money emerges as a coordination equilibrium: an asset becomes valuable because rational agents expect others to hold and accept it. High λ is the blockchain-observable, publicly verifiable signal of this coordination commitment. Second, the *convenience yield* (Sockin and Xiong, 2020; Krishnamurthy and Vissing-Jorgensen, 2012) is the non-pecuniary benefit agents derive from holding a monetary asset: access to network security, platform liquidity, and future transactional capability. Just as holders of U.S. Treasury securities accept lower yields in exchange for liquidity services (Nagel, 2016), holders of dominant blockchain coins accept lower required returns because network membership has value beyond current transactions. Crucially, the SoV premium is the present value of expected *future* convenience yields: agents who lock coins today are paying for anticipated growth in the network's membership value. Third, the *seigniorage* channel (Cong and He, 2024) anchors this signal in rational optimization: in proof-of-stake networks, newly minted tokens distributed to stakers constitute monetary seigniorage, and the staking decision reveals belief that this income plus anticipated price appreciation exceeds the opportunity cost of locking capital. These three channels are complementary: coordination theory explains why high λ is self-reinforcing, convenience yield provides the holding motive, and seigniorage grounds the rational optimization.

Governance and utility tokens present a related but distinct case. A governance token such as UNI is the monetary instrument of the Uniswap protocol economy. Its economic output is the total throughput of the Uniswap protocol—decentralized exchange volume, liquidity deployed, fees generated. Whether protocol revenue currently accrues to token holders is immaterial, just as it is immaterial whether U.S. GDP flows to dollar holders. This framing aligns with Cong, Li, and Wang (2021), who price tokens on the present value of future platform utility and adoption, and Schilling and Uhlig (2019), who model the value of digital currencies as functions of the economic activity they support. For governance tokens, λ carries a different signal. Staking a governance token into a DAO contract renders it illiquid: staked tokens cannot be sold while locked. An agent who voluntarily incurs this illiquidity cost reveals a preference for future appreciation over present liquidity—a choice grounded in assessments of network effects, utility expansion, competitive positioning, and innovation trajectory. Drawing on Hirschman (1970), holders who exercise *voice* (governance participation) rather than *exit* (selling) reveal a preference for the protocol's future that passive buyers do not. Edmans and Manso (2011) formalize this in a corporate setting:

blockholders who engage rather than exit create governance value precisely because engagement is costly and therefore credible. Applied to tokens, active governance participation is a costly signal (Spence, 1973) of fundamental conviction in platform growth, making high participation rates an economic moat indicator.

We operationalize these ideas in two metrics. The *SoV/MoE ratio*—equal to $\lambda/(1-\lambda)$—measures the balance between forward-looking conviction and current transactional utility, and is our primary return predictor. The *NVT/fundamental ratio* normalizes market capitalization by the present value of the network's expected transaction stream (constructed analogously to a PMT-based discounted cash flow, as detailed in Section 3), yielding a measure of whether the asset is cheap or expensive relative to its fundamental economic activity. Crucially, NVT/fundamental does not carry its own independent theoretical prediction for returns. Rather, it serves as a conditioning variable: high $\lambda/(1-\lambda)$ is most valuable as a return predictor when the market has not yet priced in the underlying conviction signal—that is, when NVT/fundamental is low. Assets that combine high conviction with cheap valuation (Stars) offer the best expected returns; assets that combine low conviction with expensive valuation (Avoid) offer the worst. We find that $\lambda/(1-\lambda)$ alone positively predicts future returns cross-sectionally for both coins and governance tokens. The predictive power is substantially stronger for assets with below-median NVT/fundamental, consistent with the view that NVT identifies when the conviction signal remains unpriced. A long Stars/short Avoid calendar-time portfolio generates an annualized alpha of [X]% with a Sharpe ratio of [Y] after controlling for market, size, and momentum factors, with the results holding separately for coins and governance tokens.

Related Literature. A growing body of work attempts to value digital assets, and our paper connects to several strands. On equilibrium cryptocurrency valuation, Pagnotta and Buraschi (2022) model Bitcoin value as a function of user adoption and network security; Biais, Bisière, Bouvard, and Casamatta (2019) study equilibrium selection in blockchain protocols; Schilling and Uhlig (2019) develop a general equilibrium model of Bitcoin as a competing currency; and Easley, O'Hara, and Basu (2019) analyze how transaction fees connect mining incentives to market value. Cong, Li, and Wang (2021) and Sockin and Xiong (2020) are the closest to our approach in modeling token prices through adoption dynamics and convenience yields respectively; we complement them by providing an empirically tractable QTM decomposition and a testable investment signal derived from on-chain observables. On crypto risk factors and return predictability, Liu and Tsyvinski (2021) identify network, momentum, and investor attention factors; Liu, Tsyvinski, and Wu (2022) develop a three-factor model for cryptocurrency returns; and Makarov and Schoar (2020) document price dislocations across exchanges. We contribute to this literature the first on-chain supply-side signal—$\lambda/(1-\lambda)$—grounded in monetary theory rather than return-based factor construction. On the monetary economics of digital assets, Kiyotaki and Wright (1993), Lagos and Wright (2005), and Schilling and Uhlig (2019) provide search-theoretic and general equilibrium foundations; Cong and He (2024) analyzes proof-of-stake tokenomics; and Jiang, Krishnamurthy, and Lustig (2024) draw the analogy between dollar exorbitant privilege and dominant-network premia. The convenience yield literature (Krishnamurthy and Vissing-Jorgensen, 2012; Nagel, 2016) motivates the monetary premium in $\mathrm{MC}_{\mathrm{SoV}}$. On token issuance and governance, Howell, Niessner, and Yermack (2020) study the determinants of ICO success; Chod and Lyandres (2021) model ICO design under information asymmetry; and Edmans and Manso (2011), Hirschman (1970), and DAO Governance (2024) provide the governance theory and evidence linking participation to token value. Our paper is the first, to our knowledge, to derive a directly observable valuation decomposition from QTM, to distinguish the signal content of λ for coins versus governance tokens, and to demonstrate that the interaction of conviction and valuation predicts cross-sectional cryptocurrency returns.

The remainder of the paper is organized as follows. Section 2 develops the theoretical framework and states the paper's testable hypotheses. Section 3 describes the data and variable measurement. Section 4 presents the main results. Section 5 reports robustness tests. Section 6 concludes.

2 Theoretical Framework and Hypothesis Development

The QTM decomposition in Appendix A is an accounting identity: given λ, the SoV/MoE ratio follows mechanically. It is not, by itself, a behavioral prediction. This section asks why λ, and its interaction with valuation, should predict future returns. We develop three stylized one-period models—one for the coin staking decision, one for governance-token signaling, and one for the price-discovery role of NVT/fundamental—each built directly on the theoretical traditions introduced in Section 1. Each model yields a proposition, stated formally as a hypothesis, that we take to the data in Section 4.

2.1 The Staking Decision for Coins

Consider a continuum of agents, indexed by $i \in [0,1]$, who hold the native coin of an L1 network at date t and choose whether to lock it for one period (stake) or hold it unstaked. Staking pays a seigniorage return $s_t \geq 0$—newly issued tokens distributed to validators—and a convenience yield realized at $t+1$, the non-pecuniary benefit of network membership emphasized by Sockin and Xiong (2020): validation rights, governance access, and anticipated future transactional capability. We decompose agent i's convenience yield into a common component θ_{t+1}—the network's true future growth trajectory, unobserved at t—and an idiosyncratic, mean-zero noise term ε_i reflecting agent i's imperfect private read on that trajectory: $c_{i,t+1} = \theta_{t+1} + \varepsilon_i$, with ε_i i.i.d. across agents and independent of θ_{t+1}. Staking also imposes a private liquidity cost φ_i, i.i.d. across agents and independent of $(\theta_{t+1}, \varepsilon_i)$, reflecting each agent's individual demand for transactional flexibility. Agent i stakes if and only if

$$s_t + \theta_{t+1} + \varepsilon_i \geq \varphi_i \quad \Longleftrightarrow \quad \eta_i \leq s_t + \theta_{t+1}, \qquad \eta_i \equiv \varphi_i - \varepsilon_i.$$

Because η_i is i.i.d. across the continuum of agents with fixed CDF H (the convolution of the liquidity-cost distribution and the private-signal noise distribution), the strong law of large numbers implies that the realized cross-sectional locking ratio converges almost surely to

$$\lambda_t = H(s_t + \theta_{t+1}),$$

a deterministic, strictly increasing function of the network's true future growth trajectory θ_{t+1}, given the publicly observed seigniorage rate s_t. Idiosyncratic noise in individual costs and private signals washes out in the aggregate; what survives is exactly the cross-sectional average of private information about θ_{t+1}. This is the precise sense in which λ_t is a publicly verifiable signal of the holder base's aggregate conviction about the network's future, consistent with the monetary search-theoretic view of λ as a coordination signal (Kiyotaki and Wright, 1993; Lagos and Wright, 2005)—and it is a derived property of the model above, not an assumption asserted on λ_t directly.

That λ_t aggregates private information about θ_{t+1} does not imply that θ_{t+1} is already priced at t. Recovering θ_{t+1} from λ_t requires netting out the observed seigniorage rate s_t and inverting H, which in turn requires knowing the population distribution of liquidity costs and signal noise—itself not common knowledge. Only a fraction of market participants perform this extraction, so, as in Grossman and Stiglitz (1980), costly information acquisition prevents λ_t from being fully impounded into price at t. As θ_{t+1} materializes and becomes public over $t+1$, price converges toward the fundamental value already anticipated by the high-λ holder base, so realized returns are higher for assets with higher λ_t. This partial-revelation step is the one substantive assumption in the model—a standard and narrowly scoped one, not the broad assertion the earlier draft of this section placed directly on λ_t—and we test its return implication directly as Proposition 1.

Proposition 1. *$Cov(r_{t+1}, \lambda_t) > 0$: the locking ratio, and hence the SoV/MoE ratio $\lambda_t/(1-\lambda_t)$, positively predicts the cross-section of subsequent coin returns.*

This is Hypothesis 1a, restated formally as an empirical proposition:

Hypothesis 1a (Coins). *For L1 native coins, assets with higher* $\lambda/(1-\lambda)$ *in month* t *earn higher returns over months* $t+1$ *through* $t+h$*, after controlling for size, momentum, and market beta.*

The seigniorage channel (Cong and He, 2024) reinforces Proposition 1: because s_t enters the staking threshold directly, staking is a revealed preference that seigniorage income plus anticipated price appreciation exceeds the opportunity cost of locking capital—a rational decision rule rather than a behavioral artifact.

2.2 Costly Signaling for Governance Tokens

For governance tokens, agents hold private, binary beliefs about future protocol quality, $q_i \in \{q_H, q_L\}$ with $q_H > q_L$, informed by usage, technical engagement, or other unobservable signals. Staking for governance renders the token illiquid and requires an engagement cost $\kappa > 0$—the time and attention cost of participating in governance, i.e., exercising *voice* rather than *exit* (Hirschman, 1970; Edmans and Manso, 2011). Let Δ denote the expected price appreciation conditional on protocol success and ℓ the option value of liquidity foregone by staking. Agent i stakes if and only if

$$q_i\Delta - \ell - \kappa \geq 0 \quad \Longleftrightarrow \quad q_i \geq \bar{q} \equiv \frac{\ell + \kappa}{\Delta}.$$

Because the payoff to staking is increasing in q_i while the cost $\ell + \kappa$ is common across types, this is a single-crossing condition: for $\bar{q}$ strictly between q_L and q_H, only high-quality-belief agents find staking worthwhile, and a separating equilibrium in the spirit of Spence (1973) obtains. The observed governance staking ratio λ_t is therefore informative of the fraction of the holder base with favorable private beliefs about protocol quality.

Because q_i is unobservable and revealed only through the costly action of staking—not through costless disclosure—and because governance and voting data diffuse into prices more slowly than raw volume and price data, the market only partially impounds this information at t. Subsequent returns are therefore increasing in λ_t.

Proposition 2. *For governance tokens, $\lambda_t/(1-\lambda_t)$ positively predicts subsequent returns, because it reveals the share of the holder base with favorable private information about protocol quality not yet reflected in price.*

Hypothesis 1b (Governance Tokens). *For DeFi governance tokens, assets with higher governance staking $\lambda/(1-\lambda)$ in month t earn higher returns over months $t+1$ through $t+h$, after controlling for protocol category, size, and momentum. Voting-weighted λ has incremental predictive power beyond passive staking λ alone.*

The voting-weighted extension follows directly from the same model. Suppose active voting requires an additional monitoring cost $\psi_i > 0$ beyond the cost of staking alone (passive staking, or auto-compounding, requires no ongoing engagement). The threshold for voting, $\bar{q}' = (\ell + \kappa + \psi)/\Delta$, exceeds $\bar{q}$, so voters are a more selected, higher-quality-belief subset of stakers. Voting-weighted λ therefore carries incremental information about the holder base's average private quality belief beyond passive staking λ alone—the empirical test of the Spence costly-signaling logic embedded in the second sentence of H1b.

The cost structure $\bar{q} = (\ell+\kappa)/\Delta$ nests both institutional forms this takes in practice. For vote-escrow protocols that require an explicit lock (e.g., Curve's veCRV, Balancer's veBAL), $\ell > 0$ is a real, continuously variable illiquidity cost that scales with chosen lock duration, and locked supply is the natural empirical proxy for λ_t. For protocols whose governance is decided by simple token-balance voting with no lock (e.g., Uniswap, Compound), $\ell = 0$: liquidity is never forfeited, and κ, the cost of monitoring the protocol and casting an informed vote, is the only force sustaining the separating equilibrium. The threshold logic is identical in both cases, but the observable action revealing $q_i \geq \bar{q}$ differs: locked supply where a lock mechanism exists, and otherwise the joint behavior of holding without selling—forgoing the option to exit, in the spirit of Hirschman (1970)—and active voting participation. We therefore treat λ_t as an empirical index over whichever conviction-revealing channels are observable for a given protocol—staking/locking, holding duration, and voting engagement—with the specific construction and channel weighting determined by data availability and detailed in Section 3.

2.3 Price Discovery and the Conditioning Role of NVT/Fundamental

Sections 2.1 and 2.2 establish that λ_t carries private information not yet fully reflected in price. The remaining question is what determines *how much* of that information remains unpriced at a given date. Let $\delta_t \in [0, 1]$ denote the fraction of the conviction signal embedded in λ_t that has already been absorbed into the current price p_t, so that $p_t = p_t^{\text{fund}}(1 + \delta_t)$ for some notion of fundamental value p_t^{fund} consistent with current economic activity. Holding fundamentals fixed, a higher current price relative to current transaction-based fundamental value corresponds to a higher δ_t; this is exactly what our NVT/fundamental ratio measures, since $\text{NVT}_{f,t} = \text{MC}_t/PQ_t^*$ is increasing in price for fixed PQ_t^*. The expected return earned from t to $t + 1$ as the conviction signal is gradually validated and absorbed into price is proportional to the *unabsorbed* portion of the signal, $\lambda_t(1 - \delta_t)$, rather than to λ_t alone. It follows that the marginal predictive power of λ_t on returns is decreasing in δ_t, and therefore decreasing in $\text{NVT}_{f,t}$.

Proposition 3. *The cross-partial effect of λ_t and $NVT_{f,t}$ on expected returns is negative: the return predictability of $\lambda_t/(1 - \lambda_t)$ is concentrated among assets with low $NVT_{f,t}$.*

Hypothesis 2. *The return predictability of $\lambda/(1 - \lambda)$ documented in H1 is significantly stronger among assets with below-median NVT/fundamental ratios. Equivalently, the marginal return to a unit increase in $\lambda/(1 - \lambda)$ is higher when the asset is cheap relative to its fundamental economic activity.*

This is the central insight motivating the two-dimensional quadrant framework: $\lambda/(1-\lambda)$ supplies the conviction signal, while NVT/fundamental indicates how much of that signal the market has already absorbed. Unlike $\lambda/(1 - \lambda)$, NVT/fundamental carries no independent theoretical prediction for returns in our framework—a low NVT/fundamental ratio alone, without a high conviction signal, simply describes an asset with limited current economic activity, not necessarily one that is mispriced. Empirically, H2 implies a statistically significant interaction between $\lambda/(1-\lambda)$ and NVT/fundamental in cross-sectional return regressions, with the coefficient on $\lambda/(1-\lambda)$ larger in absolute value for below-median NVT/fundamental assets.

2.4 The Quadrant Portfolio

Propositions 1 through 3 jointly imply that the assets with the highest expected unrealized returns are those combining a strong, on-chain-verified conviction signal with valuation that has not yet absorbed it: high $\lambda/(1-\lambda)$ and low NVT/fundamental. We label this combination *Star*. The mirror-image combination—low $\lambda/(1-\lambda)$ and high NVT/fundamental, signaling both weak conviction and a valuation already stretched beyond current fundamentals—we label *Avoid*.

Hypothesis 3. *A long-short portfolio that goes long Star assets (above-median $\lambda/(1-\lambda)$, below-median NVT/fundamental) and short Avoid assets (below-median $\lambda/(1-\lambda)$, above-median NVT/fundamental), rebalanced monthly, generates positive and statistically significant alpha after controlling for market, size, and momentum factors. The Stars-minus-Avoid Sharpe ratio significantly exceeds that of equal-weighted benchmark portfolios.*

H3 is the integrating test of the framework: it asks whether the joint signal has incremental predictive power beyond either dimension of Propositions 1–3 alone. The alpha estimate quantifies the economic magnitude of the mispricing the quadrant framework identifies, net of compensation for systematic risk exposures. We report portfolio results separately for coins and governance tokens to preserve the distinction between Proposition 1's monetary-coordination mechanism and Proposition 2's costly-signaling mechanism, and test whether the alpha is larger for governance tokens, where the illiquidity cost κ creates a higher signaling barrier than the seigniorage-driven staking decision for coins.

3 Data and Variable Construction

[To be written.]

4 Results

[To be written.]

5 Robustness

[To be written.]

6 Conclusion

[To be written.]

A Derivation of the SoV/MoE Decomposition

This appendix provides the formal derivation of the SoV/MoE decomposition and its simplification to a function of λ alone.

Setup. Let M denote the total circulating supply of a digital asset and MC its market capitalization, which serves as the money supply proxy in the QTM framework. Let PQ denote nominal economic output (on-chain transaction volume for coins; protocol throughput for tokens). Standard QTM gives $\mathrm{MC} \cdot V = PQ$, so the market capitalization satisfies $\mathrm{MC} = PQ/V$.

The locking ratio. Define λ as the proportion of total supply that is locked or staked:

$$\lambda = \frac{M_{\text{locked}}}{M}, \qquad \lambda \in [0, 1).$$

The free-circulating supply is $M(1-\lambda)$. We define the *free velocity* V_{free} as the velocity of only the economically active (unlocked) portion:

$$V_{\text{free}} = \frac{PQ}{M(1-\lambda)}.$$

MoE and SoV components. The Medium of Exchange component of market capitalization is the value attributable to current transactional activity, priced at the free velocity:

$$\text{MC}_{\text{MoE}} = \frac{PQ}{V_{\text{free}}} = M(1-\lambda).$$

The Store of Value component is the residual:

$$\text{MC}_{\text{SoV}} = \text{MC} - \text{MC}_{\text{MoE}} = M - M(1-\lambda) = M\lambda.$$

The SoV/MoE ratio. The ratio of the SoV to MoE components is:

$$\frac{\text{MC}_{\text{SoV}}}{\text{MC}_{\text{MoE}}} = \frac{M\lambda}{M(1-\lambda)} = \frac{\lambda}{1-\lambda}.$$

This result shows that the SoV/MoE ratio is a monotone increasing function of λ alone, requiring no estimation of velocity, transaction volume, or any other derived quantity. As $\lambda \to 0$, the ratio approaches zero (pure medium of exchange). As $\lambda \to 1$, the ratio diverges (pure store of value).

The NVT/fundamental ratio. The Network Value to Transactions ratio is NVT $=$ MC$/PQ_{\text{annualized}}$. We construct a forward-looking version by replacing current PQ with

PQ^*, the levelized present value of the projected transaction stream:

$$PQ^* = \frac{\sum_{s=1}^{n} \frac{PQ_0(1+g)^s}{(1+r_e)^s} + \frac{PQ_0(1+g)^n(1+g_\infty)}{(r_e - g_\infty)(1+r_e)^n}}{\text{annuity factor}(r_e, n)},$$

where g is the trailing k-year CAGR of PQ, r_e is the CAPM-estimated required return, g_∞ is the terminal growth rate, and the annuity factor converts the present value to a levelized annual equivalent (analogous to the Excel PMT function). The NVT/fundamental ratio is then $\text{NVT}_f = \text{MC}/PQ^*$. Full estimation details are in Section 3.

References

Biais, B., C. Bisière, M. Bouvard, and C. Casamatta, 2019, The blockchain folk theorem, *Review of Economic Studies* 86, 1341–1371.

Chod, J., and E. Lyandres, 2021, A theory of ICOs: Diversification, agency, and information asymmetry, *Management Science* 67, 5969–5989.

Cong, L. W., Y. Li, and N. Wang, 2021, Tokenomics: Dynamic adoption and valuation, *Review of Financial Studies* 34, 1105–1155.

Cong, L. W., and Z. He, 2024, The tokenomics of staking, *NBER Working Paper* 33640.

DAO Governance: Voting power, participation, and controversy, 2024, *Distributed Ledger Technologies: Research and Practice*.

A review of DAO governance: Recent literature and emerging trends, 2025, *Journal of Corporate Finance*.

Easley, D., M. O'Hara, and S. Basu, 2019, From mining to markets: The evolution of Bitcoin transaction fees, *Journal of Financial Economics* 134, 91–109.

Edmans, A., and G. Manso, 2011, Governance through exit and voice: A theory of multiple blockholders, *Review of Financial Studies* 24, 2395–2428.

Fisher, I., 1911, *The Purchasing Power of Money*, Macmillan, New York.

Grossman, S. J., and J. E. Stiglitz, 1980, On the impossibility of informationally efficient markets, *American Economic Review* 70, 393–408.

Hirschman, A. O., 1970, *Exit, Voice, and Loyalty*, Harvard University Press, Cambridge, MA.

Howell, S. T., M. Niessner, and D. Yermack, 2020, Initial coin offerings: Financing growth with cryptocurrency token sales, *Review of Financial Studies* 33, 3925–3974.

Jiang, Z., A. Krishnamurthy, and H. Lustig, 2024, Exorbitant privilege: A safe-asset view, *NBER Working Paper* 32454.

Kiyotaki, N., and R. Wright, 1993, A search-theoretic approach to monetary economics, *American Economic Review* 83, 63–77.

Krishnamurthy, A., and A. Vissing-Jorgensen, 2012, The aggregate demand for Treasury debt, *Journal of Political Economy* 120, 233–267.

Lagos, R., and R. Wright, 2005, A unified framework for monetary theory and policy analysis, *Journal of Political Economy* 113, 463–484.

Liu, Y., and A. Tsyvinski, 2021, Risks and returns of cryptocurrency, *Review of Financial Studies* 34, 2689–2727.

Liu, Y., A. Tsyvinski, and X. Wu, 2022, Common risk factors in cryptocurrency, *Journal of Finance* 77, 1133–1177.

Makarov, I., and A. Schoar, 2020, Trading and arbitrage in cryptocurrency markets, *Journal of Financial Economics* 135, 293–319.

Nagel, S., 2016, The liquidity premium of near-money assets, *Quarterly Journal of Economics* 131, 1927–1971.

Pagnotta, E., and A. Buraschi, 2022, An equilibrium valuation of Bitcoin and decentralized network assets, *Journal of Finance* 77, 887–943.

Schilling, L., and H. Uhlig, 2019, Some simple Bitcoin economics, *Journal of Monetary Economics* 106, 16–26.

Sockin, M., and W. Xiong, 2020, A model of cryptocurrencies, *NBER Working Paper* 26816.

Spence, M., 1973, Job market signaling, *Quarterly Journal of Economics* 87, 355–374.